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Pathways to open market sustainability at Mtapa Open Market in Gweru, Zimbabwe

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Abstract

Open markets in developing countries face persistent sustainability challenges, including inadequate infrastructure, poor sanitation, and exposure to environmental stressors. This study assessed socio-economic and environmental sustainability challenges at Mtapa Open Market in Gweru, Zimbabwe. A descriptive case study design was adopted, integrating questionnaire surveys ($n=415$), semi-structured interviews, and field observations. Quantitative data were analysed using SPSS and Microsoft Excel, while qualitative data were thematically analysed to enable triangulation. Findings show that seasonal climatic variability significantly disrupts trading activities, with 65% of traders identifying the rainy season as the most challenging period due to inadequate infrastructure. Sanitation access remains limited: only 26% of traders use public toilets, while 40% rely on informal alternatives, thereby increasing environmental and public health risks. Organic waste constitutes 37% of total waste generated, reflecting the dominance of fresh produce trading. The study highlights the need for improved infrastructure, affordable sanitation services, and enhanced capacity-building, supported by inclusive governance aligned with Zimbabwe's National Development Strategy 1, Vision 2030, and Africa's Agenda 2063. The dynamic nature of informal markets implies that sustainability conditions are subject to change over time. This reveals the importance of periodic assessments and longitudinal studies to track sustainability trajectories and refine interventions over time. While context-specific, the insights generated apply to other informal markets in Zimbabwe and similar urban settings in Southern Africa, where informality and service deficits remain prevalent.

Keywords Urban sustainability, Market sustainability, Sustainability challenges, Open markets, Market trade

1 Introduction

Open markets are integral to the economic and social fabric of cities, particularly in the Global South, where they function as critical platforms for informal trade and social interaction. These markets supply affordable goods to urban populations while sustaining the livelihoods of traders and their families [6, 38, 47]. Open markets are typically located in densely populated urban areas and involve high volumes of daily transactions in both locally produced and imported goods [11, 22, 30]. Despite their significance, the



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sustainability of open markets is increasingly threatened by rapid urbanisation, environmental pressures, and widening socio-economic inequalities. Urban sustainability challenges refer to the environmental, social, and economic constraints that undermine the capacity of cities to achieve long-term, inclusive, and resilient development [37, 46]. The long-term viability of open markets is therefore closely linked to both environmental and socio-economic sustainability. Environmental sustainability relates to the protection of ecological systems through effective waste management, reduced pollution, efficient resource use, and safeguarding public health [31]. Socio-economic sustainability, on the other hand, focuses on equitable economic opportunities, social justice, community participation, and the preservation of local cultures and livelihoods [13, 23].

In developed countries, open markets increasingly demonstrate how sustainability principles can be effectively integrated into market design, management, and governance. Markets such as the Grand Bazaar in Istanbul and Camden Market in London are permanent, well-regulated, and supported by established architectural and institutional frameworks [16, 24, 55]. Similarly, wet markets in Hong Kong, Borough Market in London, Rotterdam's Market Hall, and La Boqueria Market in Barcelona have evolved into major tourist attractions and are often cited as successful examples of urban regeneration [22]. These markets combine functional infrastructure, sanitation, waste management, and strong governance systems, demonstrating how open markets can benefit authorities, traders, and consumers alike. In Thailand, the Talad Tai Open Market in Phitsanulok further illustrates this point, as its simple yet adaptable spatial arrangement supports efficient trade while accommodating growth [39]. Collectively, these cases highlight how sustainability-oriented planning and management can enhance market performance and longevity.

Open markets in developing countries face persistent sustainability challenges, largely exacerbated by rapid urbanisation, limited resources and weak institutional support. Urban redevelopment initiatives, although often intended to modernise cities, have in some cases undermined market sustainability. In Ghana, efforts to regenerate municipal markets resulted in significant disruptions for traders in Cape Coast, who were relocated to poorly designed sites lacking essential infrastructure such as sanitation and electricity [8, 10]. Increased operating costs, reduced accessibility and the loss of established customer bases further weakened traders' economic resilience. Similarly, the Kiserian Market Centre in Kenya continues to struggle due to the absence of coherent legal, institutional and policy frameworks to guide its rapid expansion [21, 43, 44]. These cases underscore how inadequate governance and planning can compromise both the economic and social sustainability of open markets. Environmental health challenges further threaten market sustainability across much of sub-Saharan Africa. In Ethiopia, open markets are characterised by insufficient ablution facilities and poor sanitation, leading to disease outbreaks and heightened public health risks [4, 5, 19]. The exposure of food items to contaminants and bacteria has resulted in frequent foodborne illnesses, undermining consumer confidence and social sustainability.

Socio-economic exclusion also remains a major barrier to sustainable market management. In Zambia, informal traders are frequently marginalised from urban planning and policy processes, limiting their ability to influence decisions that directly affect their livelihoods [20]. Across the region, infrastructure regeneration initiatives are often marked by limited stakeholder participation, delayed communication, inadequate relocation

arrangements and postponed compensation payments ([8]; Asante, [12]). In Johannesburg, South Africa, the lack of adequate market infrastructure has forced traders to rely on nearby business parking lots for storage, intensifying challenges related to waste disposal, crime and service provision [2, 51]. In Harare, Zimbabwe, Mbare Musika market continues to suffer from poor sanitation, ineffective waste management and recurrent foodborne illnesses, all of which undermine its environmental and social sustainability [14, 54, 56, 57]. The City of Gweru reflects many of these broader regional challenges, with Mtapu Open Market emerging as one of the most affected sites.

The market faces numerous operational difficulties, including inadequate infrastructure, environmental health risks and socio-economic vulnerabilities among traders [18, 48]. Given Zimbabwe's weakened industrial base and the predominance of informality in the national economy, the legal recognition and sustainability of informal markets such as Mtapu Open Market are critical for maintaining local economic vitality. It is against this background that this study seeks to examine the environmental and socio-economic challenges affecting Mtapu Open Market in Gweru. The objective is to expose environmental concerns, including waste management and pollution, as well as socio-economic issues such as poverty, thereby elucidating the interconnection of these factors and their influence on market sustainability. This study was based on the following research questions: What are the environmental challenges associated with Mtapu Open Market? and What are the socio-economic challenges associated with Mtapu Open Market?

2 Materials and methods

2.1 Study area description

The study was conducted at Mtapu Open Market, the largest informal trading hub in the City of Gweru and the wider Midlands Province of Zimbabwe (Fig. 1). Mtapu Open Market is a mixed commercial space characterized by unplanned, densely arranged vending activities, offering a wide range of goods and services. The market attracts traders from

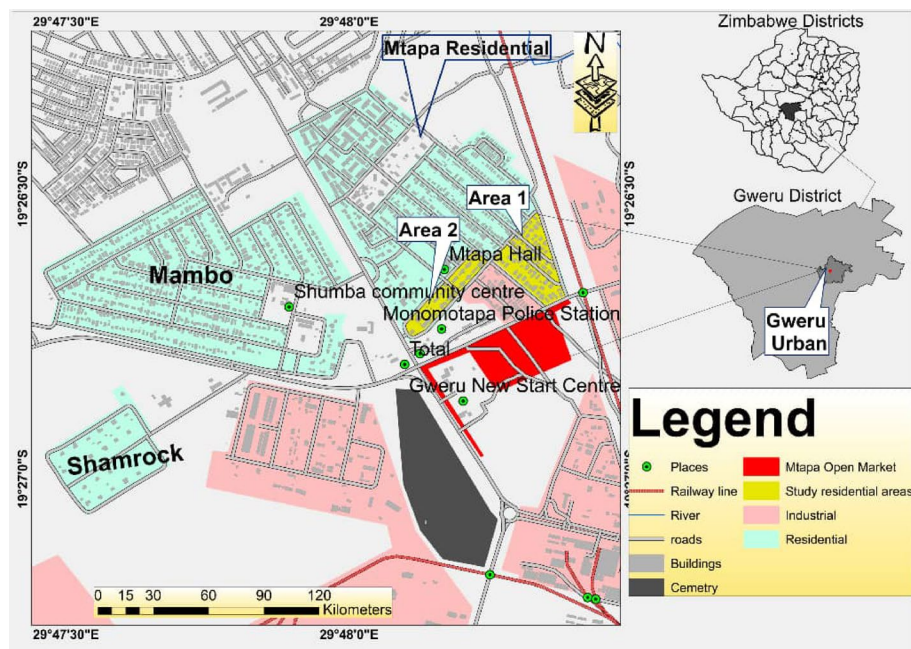


Fig. 1 Mtapu Open Market in Gweru, Zimbabwe. (Source: Authors)

Manicaland, Masvingo, and Matabeleland Provinces, making it a regional node of informal economic activity. Its emergence followed the temporary closure and partial rehabilitation of established trading spaces in Gweru in 2020, after which many traders could not afford the increased rental costs or access alternative formal markets. Consequently, Mtapá Open Market became a permanent trading space for displaced traders. The market is located on the eastern edge of the Mtapá high-density suburb, between Link Road and Matobo Road, adjacent to Monomotapa Police Station. The area is underlain by red loamy sands and black basalt soils [34] and lies within a wetland environment, which exacerbates flooding and drainage challenges during the rainy season. Climatic conditions in Gweru include a rainfall season extending from October to March or April, with mean monthly rainfall of approximately 160 mm [33], maximum summer temperatures reaching 37 °C, and winter minima falling below 6 °C [36]. These biophysical conditions make the site particularly suitable for examining environmental sustainability challenges in informal markets.

2.2 Research design and methodological rationale

A descriptive case study design was adopted to enable an in-depth examination of sustainability challenges within their real-life context. Mtapá Open Market was selected as a critical case due to its size, informality and regional importance. A mixed-methods approach was employed, integrating quantitative surveys, qualitative interviews and field observations. Methodological triangulation was considered essential in this study because sustainability challenges at informal markets are multidimensional, encompassing environmental, socio-economic and governance dimensions that cannot be adequately captured using a single method. The use of multiple data sources enhanced the credibility and robustness of findings through cross-validation of evidence.

2.3 Data collection methods

Two structured questionnaire surveys were administered: one targeting market traders and the other targeting households in Mtapá Sections 1 and 2, located adjacent to the market. Traders were selected to provide first-hand insights into trading practices, service access and operational challenges, while nearby households were included to capture external environmental and social impacts associated with market activities. Questionnaire design was guided by the study objectives, preliminary informal interviews with traders and a review of relevant literature on informal market sustainability. The questionnaires combined closed-ended and open-ended questions. Quantitative questions focused on measurable behaviours and service access (waste disposal practices, water availability, and sanitation use), enabling statistical summarisation and comparison across groups. Open-ended questions allowed respondents to elaborate on experiences and perceptions, which was particularly important for understanding complex sustainability issues. Questionnaires were self-administered to minimise interviewer bias and reduce response errors.

2.4 Sampling strategy and representativeness

The total study population comprised 703 market traders and 268 households in Mtapá Section 1 and 2, yielding a combined population of 971. Sample size determination was guided by Yamane's (1967) formula at a 95% confidence level and a 5% margin of error:

$$n = N / \left[1 + N(e)^2 \right]$$

This resulted in a total sample size of 415 respondents, comprising 255 traders and 160 household respondents. Stratified random sampling was employed to ensure proportional representation of different trader categories and household sections (Table 1). Traders were stratified according to the type of goods sold, reflecting the functional diversity of the market, while households were stratified by residential section.

2.5 Key informant interviews

Semi-structured interviews were conducted with purposively selected key informants, including officials from the Gweru City Council, the Environmental Management Agency, local political leadership, the Mtapá Open Market Vendors Association and a community-based organisation. Purposive sampling was justified by the need to obtain expert and institutional perspectives on market governance, environmental management and service provision. Interview guides were designed to explore policy implementation, institutional constraints and stakeholder roles in market sustainability.

2.6 Data analysis

Quantitative data were cleaned, coded and analysed using the Statistical Package for Social Sciences (SPSS) version 25.0 and Microsoft Excel. Descriptive statistics, including frequencies and percentages, were used to summarise survey responses. Qualitative data from interviews and open-ended questionnaire responses were analysed thematically, allowing for identification of recurrent patterns and relationships across stakeholder perspectives. Field observations were documented using photographs to support and contextualise survey and interview findings.

2.7 Ethical considerations

Ethical approval for the study was obtained from the Midlands State University Ethics Committee, Zimbabwe (Approval no. MSUFCS370625). Participation was voluntary, informed consent was obtained from all respondents, and confidentiality and anonymity were maintained throughout the research process. The methodological approach used in this paper is summarized in Fig. 2.

Table 1 Sampling procedure. *Source:* Field survey (2025)

Methodology subgroup	Population size	Grouping of trade items and respondents	Required sample size using Tara Yamane formula = $\frac{N}{1+N(e)^2}$	Sample size to represent each stratum $\frac{RequiredSample\ Size}{Population size} \times stratasize$
Market traders	703	Fruits and vegetables (244), Second-hand clothes (151), Groceries (117), New clothes (104), Hardware and packaging (62), Dried foods and grain (25)	255	Fruits and vegetables (89), Second-hand clothes (55), Groceries (42), New clothes (38), Hardware and packaging (22), Dried foods and grain (9)
Mtapá Sections 1 and 2 households	268	Mtapá Section 1(160), Mtapá Section 2(108)	160	Mtapá Section 1(96), Mtapá Section 2(64)
Total	971	971	415	415

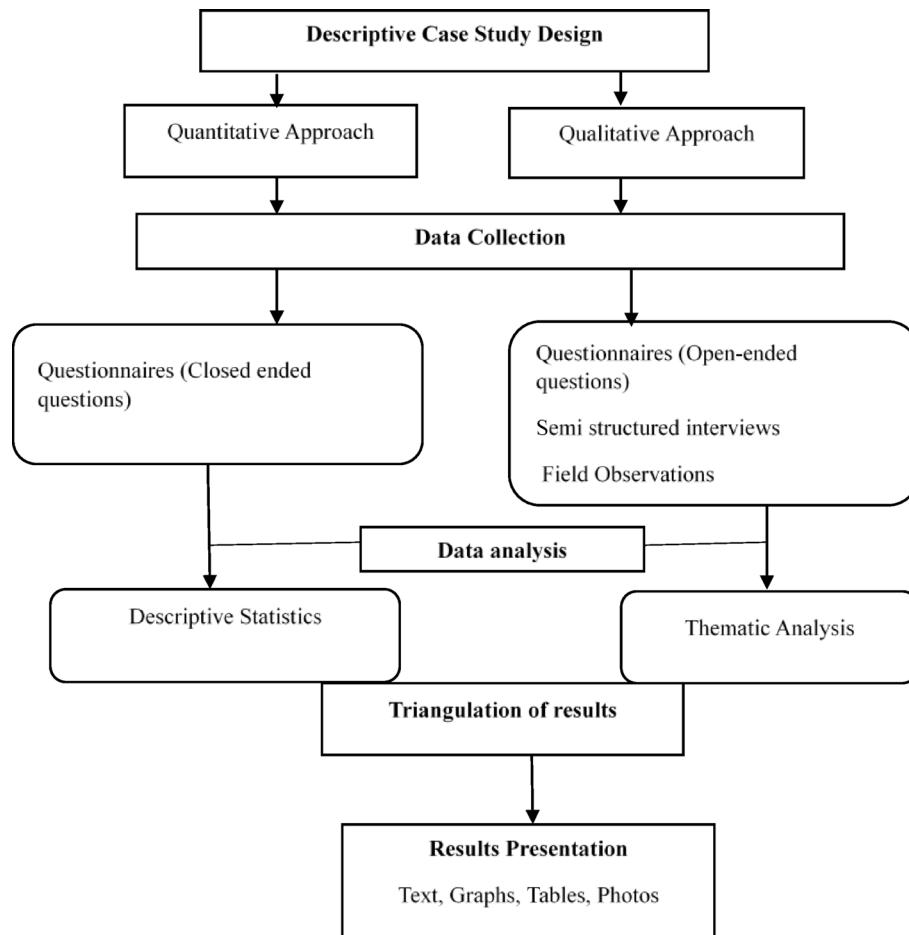


Fig. 2 Research methodology flow diagram summary

3 Results

3.1 Environmental challenges associated with the Mtapu Open Market

3.1.1 Seasonal and climatic constraints

Results indicate that seasonal variability significantly affects the sustainability and daily functioning of Mtapu Open Market. A majority of respondents (65%) identified the rainy season (late November to March) as the most challenging period for vending activities, followed by the hot season (22%) and the cold season (13%) (Fig. 3).

Field observations confirmed that heavy and frequent rainfall overwhelms the market's inadequate drainage infrastructure, resulting in waterlogged vending spaces, damage to goods, and reduced customer movement. During this period, internal market routes become muddy and slippery, increasing the risk of accidents and discouraging patronage (Fig. 4). These findings suggest that climatic impacts are not episodic but structurally embedded in the market's operational environment due to the absence of climate-resilient infrastructure.

3.1.2 Solid waste generation and management

Solid waste management emerged as the most persistent environmental challenge at Mtapu Open Market (Fig. 5). Results indicated that organic waste, primarily from fresh produce trading, constitutes the largest share of waste generated (37%), followed

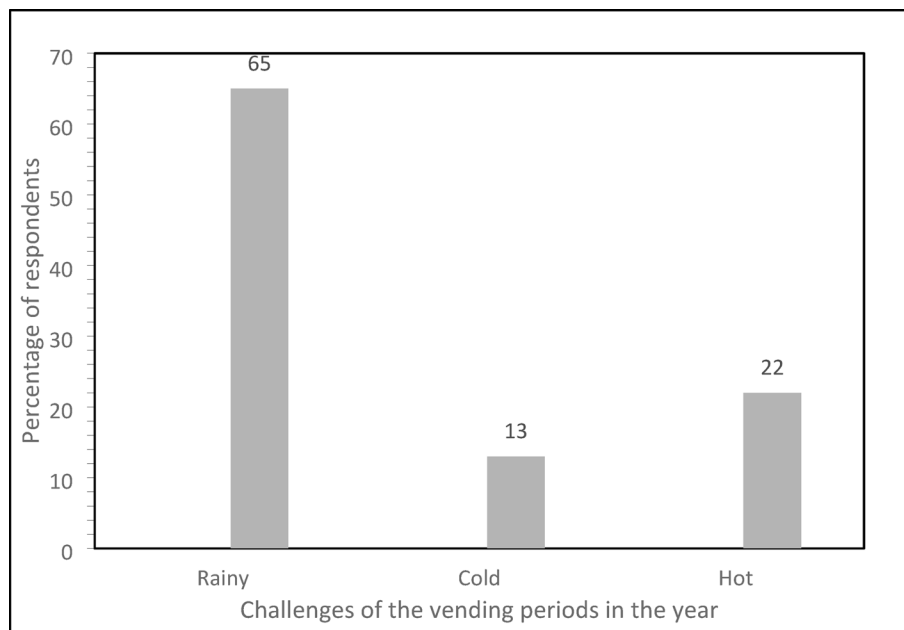


Fig. 3 Challenges of the vending periods in the year at the Mtapu Open Market



Fig. 4 Muddy roads during the rainy season at Mtapu Open Market

by plastic (22%) and metal waste (16%). Smaller proportions of respondents identified paper, fabric, human and glass waste as additional waste streams.

The households in Mtapu Sections 1 and 2 confirmed that there has been an increase in illegal dumpsites since the emergence of the market approximately 5 years ago. The households reported that solid waste from the market is seen in the nearby river, residential areas and the market area itself. The presence of ubiquitous dumpsites results in air pollution as traders burn waste after clearing vending spaces (Fig. 6).

Observations revealed that waste bins were woefully insufficient for the amount of garbage generated at the market. The GCC Cleansing Superintendent remarked, *‘The City Council has designated two skip bins at the market, each with a capacity of at least*

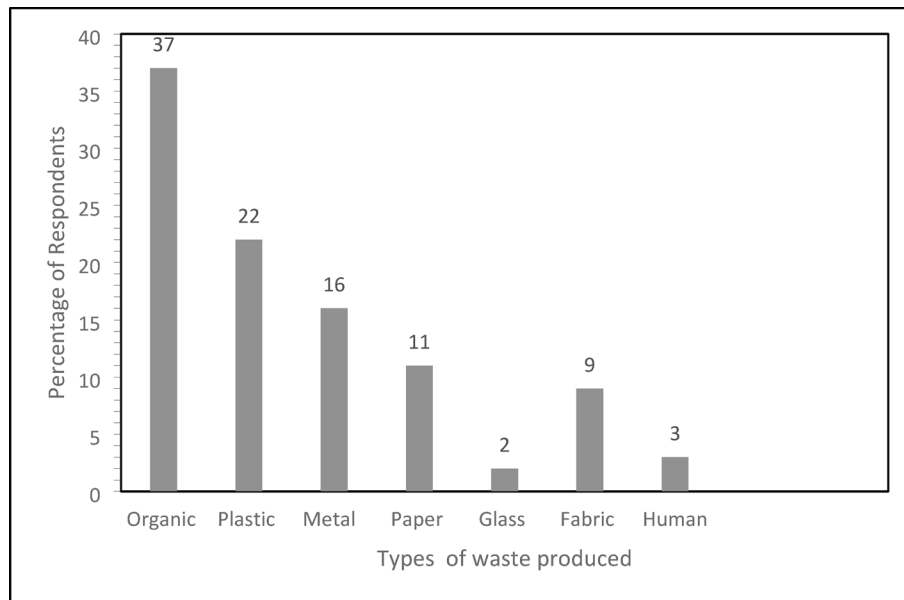


Fig. 5 Types of waste produced at Mtapu Open Market

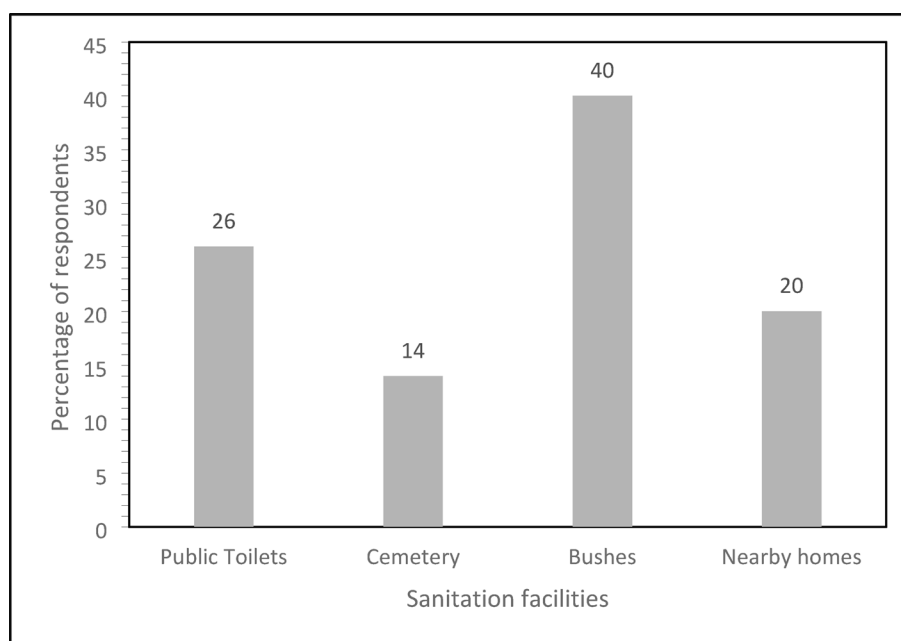


Fig. 6 Illegal waste dumpsites at Mtapu Open Market

1.2 tonnes of waste; however, this is inadequate due to the volumes of waste generated. The quantity of waste bins offered is insufficient to accommodate the daily volume of refuse created by traders and customers, causing overflow and littering throughout the market. Furthermore, the existing bins are of poor quality, frequently damaged or lacking suitable lids, attracting pests while also emitting unpleasant odours that detract from the market environment. The lack of waste segregation exacerbates the problem; without separate containers for organic, recyclable, and non-recyclable items, waste is frequently intermingled, complicating recycling efforts and increasing environmental risks (Table 2).

Table 2 Summary of key environmental challenges at Mtapu Open Market

Environmental challenge	Evidence from survey (%)	Stakeholder/field evidence	Observed impacts
Seasonal rainfall impacts	65% affected during the rainy season	Poor drainage; muddy internal roads	Damage to goods, reduced customer access, and safety risks
Dominance of organic waste	37%	Vendors Association Chairperson: <i>"The fresh produce traders are the main producers of waste at the market."</i>	Rapid decomposition, odours, pest infestation
Plastic and metal waste	22% (plastic); 16% (metal)	Litter observed within and around the market	Blocked drainage, visual pollution
Inadequate waste infrastructure	Interview feedback	GCC Cleansing Superintendent: <i>"Two skip bins are inadequate due to the volumes of waste generated."</i>	Overflowing bins, illegal dumping
Waste burning practices	Interview feedback	Vendors Association Chairperson: <i>"We end up burning waste because the city council fails [to collect] it timeously."</i>	Air pollution, health risks
Human waste presence	Interview feedback	Humwe Eden Project Coordinator: <i>"Human waste is present within the market environment due to traders' open defecation practices."</i>	Environmental contamination, public health risks

**Fig. 7** Sanitation facilities used at the Mtapu Open Market

3.1.3 Socio-economic challenges associated with Mtapu Open Market

During an interview, the chairperson of the Mtapu Open Market Vendors Association revealed that ablution facilities are present at the market; however, they are pay-as-you-use. The cost per use was indicated as ZWL 20, equivalent to USD 0.50 or ZAR10. This fee is expensive as specified by most market traders during the questionnaire survey, with only 26% of respondents mentioning that they use these public toilets. Most (40%) market traders indicated using the nearby bushes to relieve themselves (Fig. 7). Additionally, 20% of the study participants used domestic toilets in Mtapu Sections 1 and 2. The market is located adjacent to Mtapu Cemetery which was reported to be used as an ablution facility by 14% of the respondents. The majority of household respondents

(57%) reported charging USD 1 a day for market traders to use their toilets, while 27% of respondents claimed to provide free toilet service for the traders. 16% (16%) do not permit traders to use their facilities at all. The GCC Housing and Community Services Officer said, *'Fewer people have been using the toilets.' This is reflected by the amount of money collected compared to the number of market traders and their patrons who frequent the market daily.'*

Figure 8 illustrates the market infrastructure as observed during field observations in the study area. Researchers noted that most vending stalls at the Mtapa Open Market were built with substandard materials, including low-quality wood, plastic, and metal. The Housing and Community Services Officer remarked, *'Our department did not construct vending stalls at the market due to a lack of funds; instead, we allocated specified vending spaces to traders, granting them the freedom to design and build their compartments.'*

The majority of market respondents (93%) stated that they leave their merchandise at the market when they depart while 7% carried their products home. In an interview, the chairperson of the Mtapa Open Market Vendors Association said, *'Traders are required to pay a daily security guarding fee of USD 1 and a monthly operational fee to the council, pegged at USD 14.'* Most respondents (63%) indicated that the market fees are very high, followed by high (17%), moderate (11%) and low (9%). The Mtapa Open Market Vendors Association chairperson said, *'The market attracts traders from places as far as Manicaland, Masvingo, and Matabeleland Provinces, among other areas in Zimbabwe, who transform their workplaces into shacks at night for temporary residence'* (Fig. 9). The Ward 18 Councillor also revealed that the female traders at the market face operational challenges like issues related to personal safety, sexual harassment, exploitation and abuse. Health challenges were also mentioned as barriers that impact their ability to work. The Housing and Community Services Officer explained the informality of the market during an interview, saying, *'All traders are informal, although their presence at*



Fig. 8 Vending stalls present at Mtapa Open Market



Fig. 9 Shacks used as shelter by traders at night

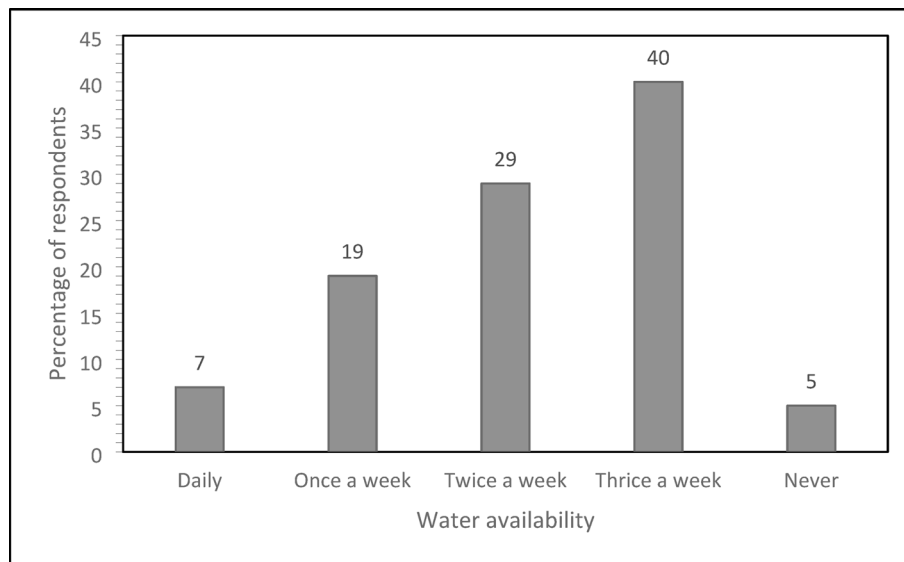


Fig. 10 Availability of water at the Mtapa Open Market

the market is known to the council. There is no proper licensing; however, the GCC provides traders with a “bin card” after paying the required monthly market fees. In this context, a bin card refers to document produced the GCC giving the traders permission to operate within council space.

Regarding the water supply schedule at the Mtapa Open Market (Fig. 10), most traders (40%) reported receiving water three days a week, followed by twice a week (29%), once a week (19%) and never (5%). On the contrary, the GCC Housing and Community Services officer specified that the water supply at the market is the same as for the households in Mtapa Sections 1 and 2. An interview with the Ward 18 Councillor said, ‘*Water is supplied in Mtapa Sections 1 and 2 from Monday evening to Thursday morning.*’ This supports the results from the market traders who indicated that they receive water at least 3 days a week. In an interview, the Humwe Eden Project Coordinator commented, ‘*The traders regularly visit our complexes to fetch water since we have a borehole to complement our water supply.*’

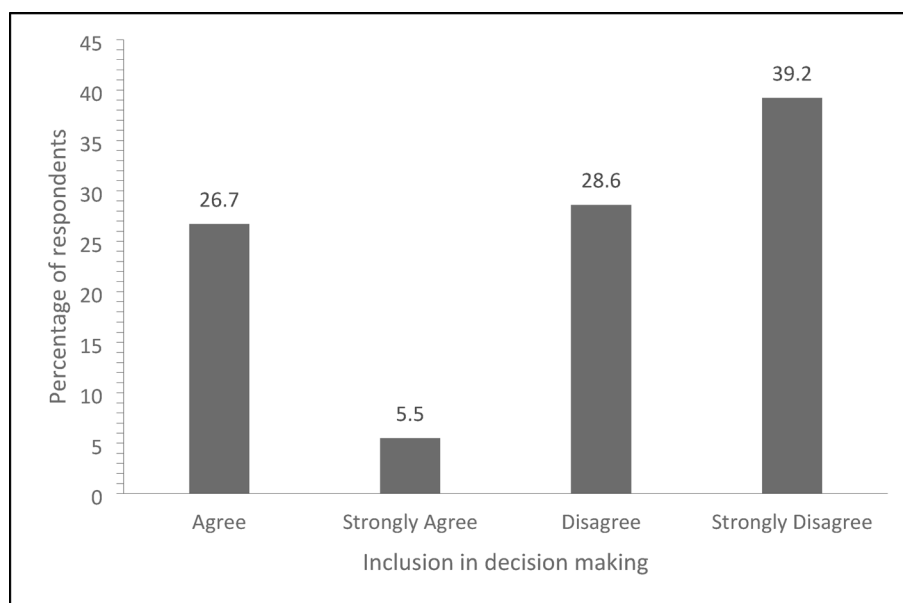


Fig. 11 Inclusion in the decision-making process at Mtapa Open Market

Table 3 Socio-economic challenges and differential impacts at Mtapa Open Market

Challenge	Most affected group(s)	Evidence	Implications
High sanitation costs	Low-income traders, women	Only 26% use public toilets; 40% use bushes	Open defecation, health risks
Informal infrastructure	All traders	GCC Housing Officer: <i>"Traders were given freedom to design and build their compartments."</i>	Exposure to weather, insecurity
Safety and harassment	Female traders	Ward 18 Councillor reported sexual harassment and abuse	Gendered vulnerability
Overnight sheltering	Long-distance traders	The Vendors Association Chairperson reported that traders sleep in shacks	Health, safety and dignity concerns
Water scarcity	Fresh produce traders	Traders fetch water from nearby boreholes	Poor hygiene, reduced productivity

Mtapa Open Market traders specified that they were not included in the decision-making process at the market, as strongly agreed by most respondents (39.2%). The study also revealed that 26.7% of the respondents agreed to being part of the decision-making process (Fig. 11). The GCC Housing and Community Services Officer confirmed, *'We regularly communicate with the Mtapa Open Market Vendors Association Committee to pass information and to get the contributions and queries of the traders.'* The Ward 18 Councillor also said, *'We are part of the committee responsible for the day-to-day functioning of the market, among other Local Authority members. This is where we receive important information for improving the functionality of the market.'*

The socio-economic challenges encountered by traders at Mtapa Open Market were summarised (Table 3) together with the implications on the affected groups observed from the study.

4 Discussion

The findings demonstrate that seasonality and climate variability exert a profound influence on the sustainability of Mtapá Open Market, shaping both operational practices and traders' livelihoods. Traders operating in open environments are directly exposed to extreme temperatures, rainfall, humidity, and wind, which reduce customer turnout, accelerate spoilage of perishable goods, and lead to immediate income losses. Given that most traders rely on daily sales to meet household needs, these disruptions have cumulative effects that threaten household economic security. This finding aligns with broader evidence from informal markets across the Global South, which shows that climate-sensitive trading environments disproportionately affect informal traders who lack protective infrastructure [52, 53]. However, the Mtapá Open Market illustrates how these global patterns manifest locally through heightened spoilage, fluctuating demand, and increased vulnerability to health risks. Environmental exposure also raises significant public health concerns. Traders and customers are susceptible to windborne and waterborne diseases, particularly during periods of heavy rainfall or extreme heat. Consistently in South Africa, such exposure contributes to respiratory, musculoskeletal, and psychological health challenges among informal traders [52]. These conditions undermine progress towards Sustainable Development Goal (SDG) 3 on good health and well-being. Findings suggest that limited municipal resources, weak institutional coordination, and competing urban priorities constrain the implementation of climate adaptation strategies for informal markets.

Solid waste management emerged as another critical environmental challenge at Mtapá Open Market, which generates diverse waste streams, with food waste dominating due to the high concentration of fresh produce vendors. Correspondingly, Jerie and Tevera [27] and Jerie [28] concurred that significant amounts of solid waste are generated in the informal sector of Gweru, particularly in market areas that focus on the retailing of vegetables and food products. Inadequate waste collection, insufficient recycling infrastructure, and irregular refuse removal have resulted in open dumping and the proliferation of illegal dumpsites. These conditions mirror challenges documented in open markets across Africa (Sloother, [54]; [56]) and reinforce evidence that informal market waste management remains marginalised within urban service delivery systems [3]. Ineffective waste management poses risks of foodborne illnesses and pest infestation and reflects missed opportunities for circular economy initiatives that could generate income and enhance market sustainability. The persistence of these challenges reflects implementation barriers like limited funding, lack of trader-municipality collaboration, and the absence of incentives for waste separation and recycling.

The study further reveals that socio-economic pressures significantly shape traders' experiences at Mtapá Open Market. Traders are required to pay fees for vending space, sanitation, and security, which many perceive as disproportionate to the quality and accessibility of services provided. Pay-to-use toilets were widely described as unaffordable, contributing to declining usage and prompting unhygienic coping practices, including open defecation in nearby undesignated areas. The use of Mtapá Cemetery for sanitation purposes is particularly a defilement of the local cultural context, where cemeteries are regarded as sacred spaces. This practice raises public health concerns and generates social and spiritual tensions, reinforcing community stigma and disrespect for the deceased [26]. Similar sanitation challenges have been reported in other Zimbabwean

markets, such as Mbare Musika, indicating a systemic failure to provide inclusive and affordable sanitation infrastructure in informal trading spaces [14, 57]. Vending infrastructure at Mtapu Open Market further reflects broader issues of underinvestment and poor urban planning. Temporary and substandard materials expose traders and customers to safety risks during storms and extreme weather events, undermining both environmental aesthetics and operational efficiency. Although durable and climate-resilient market infrastructure is widely recognised as best practice, its implementation remains weak due to fiscal constraints, limited political prioritisation of informal economies, and the ambiguous legal status of informal traders within urban governance frameworks. These findings highlight the tension between the economic importance of informal markets and their continued marginalisation in urban development strategies, with implications for SDG 8 on decent work and economic growth.

Water access is another determinant of market sustainability. Inconsistent water supply at Mtapu Open Market, linked to infrastructure breakdowns and municipal financial constraints, compromises hygiene practices and increases the risk of waterborne and foodborne diseases. Despite the market's importance in local trade, inconsistencies in water supply schedules present considerable obstacles for traders and the wider Gweru City community. Mark and Tafadzwa, [35] confirmed that equipment breakdown at the Gwenoro Water Treatment Plant and the Gweru City Council's financial incapacity to repair and maintain it result in water supply challenges in Mtapu, exposing residents and traders to unhygienic practices. This challenge reflects broader urban service delivery failures in Zimbabwe following prolonged economic decline and climate stressors. Evidence from Nigeria and Mozambique similarly shows that water scarcity in open markets contributes to widespread food contamination and public health burdens [41, 49]. Improving water distribution and reliability at Mtapu Open Market is therefore central to traders' well-being and to achieving SDG 6 on clean water and sanitation.

Beyond environmental and infrastructural constraints, the findings underscore significant governance and social inclusion challenges. Most traders reported exclusion from decision-making processes, indicating weak communication channels between the Gweru City Council and market stakeholders. This lack of participatory governance undermines social sustainability and echoes experiences from other regions, where informal traders are marginalised from urban planning despite their economic contribution [17, 20, 29]. At Mtapu Open Market, the absence of inclusive decision-making limits traders' ability to influence service provision, infrastructure development, and environmental management, perpetuating cycles of neglect. The study also highlights important intersectional dimensions, particularly gender, whereby women often enter the sector out of economic necessity to support households and care for children. Nursing mothers face compounded challenges as they juggle childcare and vending responsibilities in environments lacking adequate sanitation, water, and shelter. Consistent with evidence from Arba Minch city, Ethiopia, women in informal markets experience heightened exposure to health and safety risks compared to men ([1, 7]; Njeru et al. 40). These gendered vulnerabilities demonstrate that uniform policy responses are insufficient; instead, market interventions must account for intersecting factors such as gender, caregiving roles, age, and physical ability.

4.1 Opportunities for sustainability at Mtapu Open Market, Gweru, Zimbabwe

Open markets in urban contexts play a critical role in fostering social inclusion and enhancing urban vibrancy through face-to-face trading practices that provide authentic economic and cultural experiences. Such markets are not only central to the livelihoods of low-income urban populations but also serve as key drivers of local economic development by generating employment, income, and municipal revenue. In Gweru, Mtapu Open Market exemplifies the socio-economic importance of informal trading, while simultaneously revealing sustainability challenges that undermine its long-term viability. Figure 12 highlights the principal environmental, social, and infrastructural challenges identified in the study, alongside opportunities for advancing sustainability within the market.

Given the substantial contribution of informal trading to household livelihoods and the broader urban economy, there is an urgent need to improve the operational conditions at Mtapu Open Market in a manner that is both inclusive and sustainable. Enhancing market performance requires the systematic improvement of operational procedures through participatory engagement with key stakeholders, including traders, municipal authorities, public health officials, and community leaders. Such engagement can facilitate the formulation and implementation of context-specific policies and interventions that support market growth while safeguarding public health, environmental quality, and cultural heritage. One of the most immediate priorities is the improvement of sanitation and waste management systems to protect the health and safety of traders and consumers. The informal nature of the market currently compromises food safety and contributes to environmental degradation, including illegal dumping and unsanitary conditions. In the short term, the City of Gweru, in collaboration with trader associations, should implement basic sanitation measures such as regular waste collection, provision of refuse bins, and enforcement of hygiene standards. In the medium term, community-led awareness campaigns and training programmes can educate traders on proper waste handling, food hygiene, and environmental stewardship, thereby fostering a culture of shared responsibility and compliance.

Infrastructure development represents a medium- to long-term opportunity for enhancing market sustainability. The provision of reliable and safe drinking water, improved drainage systems, and sheltered trading spaces would significantly reduce disease outbreaks and improve working conditions. Drilling boreholes offers a practical and sustainable solution to irregular municipal water supply, ensuring consistent access to water for sanitation and trading activities. However, the implementation of such infrastructure projects may face financial constraints and institutional capacity limitations, necessitating partnerships with national government agencies, non-governmental organisations, and development partners.

Environmental sustainability initiatives, such as recycling and composting programmes, present further opportunities to improve market conditions while reducing environmental impacts. Recycling initiatives can divert waste from illegal dumpsites, lower greenhouse gas emissions, and promote a cleaner market environment. Composting organic waste generated from fresh produce trading can produce nutrient-rich compost for urban agriculture, contributing to local food security and soil fertility. These initiatives are best implemented incrementally, beginning with pilot projects supported by the municipality and civil society organisations, and scaled up as capacity

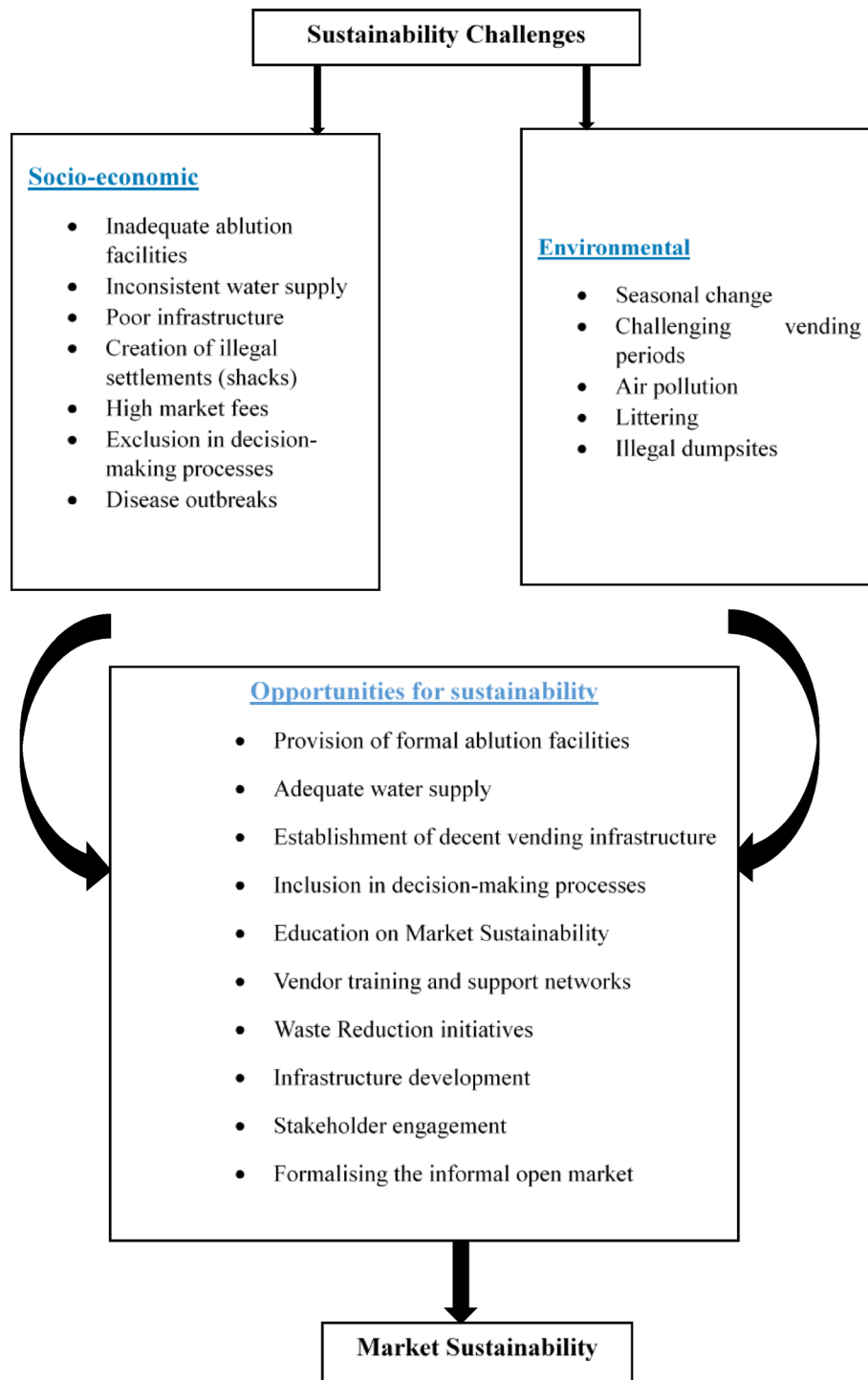


Fig. 12 Sustainability challenges associated with Mtapa Open Market

and resources allow. Despite the clear benefits of these interventions, several barriers may hinder their implementation. Financial limitations, weak regulatory enforcement, political will, and resistance to behavioural change among traders may slow progress. Addressing these challenges requires transparent governance structures, inclusive decision-making processes, and the mobilisation of both public and private resources.

Incentive-based approaches, such as reduced trading fees for compliant vendors, may also encourage adoption of sustainable practices.

Collectively, the proposed interventions provide a structured pathway for enhancing the sustainability of Mtapá Open Market. By aligning short, medium, and long-term actions with clearly defined institutional responsibilities, these measures support Zimbabwe's National Development Strategy 1 objectives on environmental protection and economic development. Furthermore, the findings contribute to the realisation of Vision 2030, which aspires to transform Zimbabwe into a prosperous and empowered upper-middle-income society starting with the Mtapá Open Market in the City of Gweru. The recommendations are also consistent with global and regional development frameworks, including the United Nations Sustainable Development Goals and Africa's Agenda 2063, particularly SDGs 1, 2, 3, 6, 8, 9, and 11 on no poverty; zero hunger; good health and well-being; clean water and sanitation; decent work and economic growth; industry, innovation and infrastructure; and sustainable cities and communities, respectively. Strengthening informal market sustainability, therefore, improves local livelihoods and also advances broader national and international development goals.

5 Conclusion

Mtapá Open Market serves as a microcosm of the intricate relationship between economic opportunity, regulatory challenges, and social dynamics within informal trading environments. This study examined the sustainability challenges facing Mtapá Open Market, focusing on environmental conditions, socio-economic vulnerabilities and governance arrangements. The findings show that environmental sustainability is constrained by seasonal flooding, inadequate drainage, poor waste management and limited sanitation facilities, challenges that are intensified by the market's wetland location and high volumes of organic waste. Socio-economically, traders operate under conditions of infrastructural inadequacy, inconsistent water supply and high operational costs, with female traders facing disproportionate risks related to safety, health and harassment. Governance analysis revealed that institutional capacity constraints and limited trader participation in decision-making processes hinder effective implementation of sustainability practices, despite general awareness of best practices. Findings highlight the need for integrated and participatory approaches to informal market management. The study demonstrates that sustainability interventions must simultaneously address infrastructure provision, service delivery, trader empowerment and governance reform. While context-specific, the insights generated apply to other informal markets in Zimbabwe and similar urban settings in Southern Africa, where informality and service deficits remain prevalent. Aligning market interventions with national development priorities and global sustainability frameworks is critical for achieving scalable and long-term impact.

5.1 Limitations of the study

This study recognises several limitations that have implications for the interpretation and application of its conclusions. First, the focus on Mtapá Open Market as a single case study means that the findings should not be interpreted as statistically representative of all open markets in Zimbabwe or beyond. Instead, the conclusions are analytically generalisable, offering insights relevant to markets with similar characteristics, such

as informality, limited infrastructure and weak institutional support. This contextual specificity implies that policy recommendations should be adapted to local conditions rather than applied uniformly. The reliance on qualitative interviews and field observations introduces the possibility of subjective bias, as stakeholder accounts reflect individual perceptions and experiences. While triangulation with quantitative survey data and multiple stakeholder perspectives was employed to enhance credibility, the conclusions may still emphasise perceived challenges over unobserved or systemic dynamics. Third, the use of selected data-collection methods may have limited the study's ability to capture broader structural influences, such as national economic policies, macro-level governance dynamics or deeply embedded cultural practices affecting market operations. As a result, the conclusions primarily address local-level sustainability challenges and institutional arrangements, suggesting the need for future research that integrates macroeconomic and policy analysis to provide a more comprehensive understanding. The dynamic nature of informal markets implies that sustainability conditions are subject to change over time. The conclusions, therefore, represent a snapshot of prevailing challenges and opportunities at the time of the study, and their relevance may evolve as market conditions, policies and infrastructure interventions change. This highlights the importance of periodic assessments and longitudinal studies to track sustainability trajectories and refine interventions over time.

Supplementary Information

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Supplementary Material 1.

Author contributions

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

Data availability

Data will be made available upon reasonable request.

Declarations

Ethics approval and consent to participate

Permission to carry out the study was sought and approved by the Gweru City Council. The authors submitted the manuscript with permission from the Faculty of Social Sciences at Midlands State University. Ethical approval for the study was obtained from the Midlands State University Ethics Committee, Zimbabwe (Approval no. MSUFCS370625). The study adheres to the ethical guidelines set forth by the Declaration of Helsinki and Belmont Report. This study was conducted under ethical guidelines and approved by the ethics committee of Midlands State University. Informed consent was obtained from all participants before their involvement in the study. Participants were informed about the study's purpose, procedures, potential risks, and benefits and their confidentiality was assured. Participation was voluntary, and they could withdraw at any time without consequence. Informed consent was obtained from all participants before their involvement in the study. Participants were advised that participation was voluntary, and they could withdraw at any time without consequence.

Consent for publication

We affirm that informed consent for publication of images of human participants included in this study has been obtained from each individual. This consent covers the use of identifiable images in the manuscript. All reasonable measures have been taken to protect anonymity and we confirm that no identifiable information is disclosed without consent.

Competing interests

The authors declare no competing interests.

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References

1. Abebe YM. Challenges and opportunities of women participating in informal sector in Ethiopia: a special focus on women street vendors in Arba Minch City. *Int J Sociol Anthropol*. 2017;9(2):8–16.
2. Adhikari S, Struwig M. Concentrations and health risks of selected elements in leafy vegetables: a comparison between roadside open-air markets and large stores in Johannesburg, South Africa. *Environ Monit Assess*. 2024;196(2):1–17.
3. Amato C, Jerie S, Mutekwa TV, Shabani T, Shabani T, Tafadzwa M. Impacts of solid waste management strategies in urban high-density suburbs: a case of Amaveni suburb, Kwekwe, Zimbabwe. *Discover Environ*. 2024;2(1):97.
4. Amenu K, Bedasa M, Wamile M, Worku H, Kasim K, Taha M, Mego L, Dinede G, Ssemenda JN, Grace D, Roesel K. Qualitative assessment of chicken and vegetable value chains in Harar and Dire Dawa, Ethiopia: Food safety perspectives. ILRI Research Report; 2021.
5. Amenu K, Megersa B, Jaleta MB, Dinede G, Worku H, Kasim K, Taha M, Ibrahim AM, Kedir J, Mego L, Roesel K. Potential food safety risks in tomato value chains in urban settings of Eastern Ethiopia: a qualitative investigation. *Front Sustainable Food Syst*. 2023;7:1–11.
6. Amoah ST. Present and active: unpacking the negotiated logics of container street traders in the governance of a Ghanaian City. 2022.
7. Amoako J. Women's occupational health and safety in the informal economy. Ghana: Maternal market traders in Accra; 2019.
8. Asante LA, Helbrecht I. Conceptualizing marketplaces in Anglophone West Africa: a sexpartite framework. *GeoJournal*. 2020;85:221–36.
9. Asante LA, Helbrecht I. Urban governance and its implications for the micro-geographies of market trading in Ghana: a case of the Kotokuraba Market Project in Cape Coast. *GeoJournal*. 2020;85(5):1203–25.
10. Asante LA, Helbrecht I. Hybrid entrepreneurial urban governance in post-colonial Ghana: an analysis of Chinese funding of the Kotokuraba market project in Cape Coast. *Urban Geogr*. 2022;43(10):1519–43.
11. Asante LA, Mills RO. Exploring the socio-economic impact of the COVID-19 pandemic in marketplaces in urban Ghana. *Afr Spectr*. 2020;55(2):170–81.
12. Asante LA. The politics and activism of urban governance in Ghana: analyzing the processes of market redevelopment in Kumasi and Cape Coast. 2020.
13. Chauhan N. Empowering communities and preserving heritage: case studies on socially responsible travel through fair trade tourism. *Meaningful tourism: strategies and future development*. Emerald Publishing Limited; 2025. pp. 163–82.
14. Chikulo S, Hebinck P, Kinsey B. Mbare Musika is ours: an analysis of a fresh produce market in Zimbabwe. *Afr Affairs*. 2020;119(476):311–37.
15. Claybaugh CC, Haried P, Chen Y, Chen L. 2021. ERP vendor satisfaction: from communication and IT capability perspectives. *J Comput Inform Syst*.
16. Dana LP, Etemad H. The bazaar economy. In: *Understanding contexts of business in Western Asia: Land of Bazaars and High-Tech Booms*. 2022. pp. 19–39.
17. Deore P, Lathia S. Streets as public spaces: lessons from street vending in Ahmedabad, India. *Urban Plann*. 2019;4(2):138–53.
18. Dzawanda B, Maderera TL, Matsa MM. Prospects for circular economy adoption in an Urban open market: case study of Mtapu Open Market, Gweru. *Circ Econ Sustain*. 2024;1–24.
19. Gemedu BA, Amenu K, Girma S, Grace D, Srinivasan R, Roothaert R, Knight-Jones TJ. Knowledge, attitude, and practice of tomato retailers towards hygiene and food safety in Harar and Dire Dawa, Ethiopia. *Food Control*. 2023;145:109441.
20. Giroux S, Blekking J, Waldman K, Resnick D, Fobi D. Informal vendors and food systems planning in an emerging African city. *Food Policy*. 2021;103:p101997.
21. Gitau FP. 2011. Urban management challenges facing emerging towns in Kenya: a case study of Kiserian Market Centre in Kajiado County (Doctoral dissertation, University of Nairobi, Kenya).
22. Gonzalez S. Contested marketplaces: retail spaces at the global urban margins. *Prog Hum Geogr*. 2020;44(5):877–97.
23. Hariram NP, Mekha KB, Suganthan V, Sudhakar K. Sustainability: an integrated socio-economic-environmental model to address sustainable development and sustainability. *Sustainability*. 2023;15(13):10682.
24. Hiebert D, Rath J, Vertovec S. Urban markets and diversity: towards a research agenda. *Cities Divers Ethn*. 2017;17–32.
25. Hkoma MB, Ashokri H, Fahej M, Bani M, Alhanbli A, Amer I, Maiteeq F. Towards green markets: strategies for enhancing sustainability in open-air popular markets, Zliten, Libya. *AlQalam J Med Appl Sci*. 2025;1477–86.
26. Hove R, Baloyi ME, Phalatsi-Shilubana M. A Critical Analysis of Cremation Burials Within Some of the South African Tribes: A contextual practical theological view. *Religions*. 2024;15(10):1264.
27. Jerie S, Tevera D. Solid waste management practices in the informal sector of Gweru, Zimbabwe. *J waste Manage*. 2014;2014(1):148248.
28. Jerie S. Occupational risks associated with solid waste management in the informal sector of Gweru, Zimbabwe. *J Environ Public Health*. 2016;2016(1):9024160.
29. Keen M, Ride A. Trading places: Inclusive cities and market vending in the Pacific Islands. *Asia Pac Viewp*. 2019;60(3):239–51.
30. Kitonyi HC. Assessing the functionality of Open-air Markets: a case of Gikomba Market, Nairobi County (Doctoral dissertation, University of Nairobi). 2023.
31. Kolawole AS, Iyola AO. Environmental pollution: threats, impact on biodiversity, and protection strategies. *Sustainable utilization and conservation of Africa's biological resources and environment*. Singapore: Springer Nature Singapore; 2023. pp. 377–409.
32. Kopczyńska E. Economies of acquaintances: Social relations during shopping at food markets and in consumers' food cooperatives. *East Eur Politics Soc*. 2017;31(3):637–58.
33. Makuvu V, Murewi CT, Dimes J, Chagonda I. Are smallholder farmers' perceptions of climate variability and change supported by climate records? A case study of lower Gweru in semiarid central Zimbabwe. *Weather Climate Soc*. 2018;10(1):35–49.
34. Matsa M, Mutekwa T. Property rights and environmental conflicts in Africa: an exploration of the main issues. *Ethiopian J Environmen Stud Manage*. 2009;2(1).

35. Mark M, Tafadzwa T. The domestic water supply puzzle in Zimbabwean cities: challenges and opportunities for Mtapa Residential Suburb, Gweru, Zimbabwe. *Int J Humanit Social Sci.* 2015;5(1):123–33.
36. Matsa MM, Chokuda F, Mupepi O, Dzawanda B. An assessment of groundwater quality in Zimbabwe's urban areas: case of Mkoba 19 suburb, Gweru. *Environmen Monitor Assessm.* 2021;193(7):391.
37. Mensah AA, Amponsah O, Takyi SA, Mensah H. Land governance in Ghana and its implications for inclusive city development. *SN Social Sci.* 2022;2(1):5.
38. Mwangi IP. Potential for periodic markets as nodes of food trade and territorial development: a case of Machakos-Kitui Road, Machakos County, Kenya (Doctoral dissertation, University of Nairobi). 2022.
39. Navapan N, Charoenkit S. Local markets: how the ordinary public places can support urban sustainable development. *GMSARN Int J.* 2022;16:308–13.
40. Njeru P. The influence of open-air market Crèche on exclusive breastfeeding practices among women in informal employment in Chuka Market, Kenya (Doctoral dissertation, University of Nairobi). 2024.
41. Nordhagen S, Onuigbo-Chatta N, Lambertini E, Wennedt A, Okoruwa A. Perspectives on food safety across traditional market supply chains in Nigeria. *Food Human.* 2023;1:333–342.
42. Obara OM. Survival of the urban poor by the open-air market at Gaba Landing Site in Kampala Environment. 2013.
43. Owuor S. HCP report no. 6: the urban food system of Nairobi, Kenya. 2018.
44. Owuor S, Brown A, Crush J, Frayne B, Wagner J. No. 06: the urban food system of Nairobi, Kenya. 2017.
45. Petrović MD, Ledesma E, Morales A, Radovanović MM, Denda S. The analysis of local marketplace business on the selected urban case—Problems and perspectives. *Sustainability.* 2021;13(6):3446.
46. Quaye I, Amponsah O, Azunre GA, Takyi SA, Braimah I. A review of experimental informal urbanism initiatives and their implications for sub-Saharan Africa's sustainable cities' agenda. *Sustain Cities Soc.* 2022;83:103938.
47. Reazul Haque M, Rahman A, April. Vegetable Street Vending in Dhaka, Bangladesh: Reflecting a Socioeconomic Perspective Toward the Informal Sector of African Entrepreneurship. In: International conference on current business issues in African Countries. Cham: Springer Nature Switzerland; 2023. pp. 217–240.
48. Runokunda JC, Matsa M, Dzawanda B, Mupepi O. Unmasking open market informality: merchandise and market interactions at Mtapa Open Market in Gweru, Zimbabwe. *Discover Global Soc.* 2025;3(1):1–21.
49. Salamandane AR. Antibiotic resistance and virulence encoding genes are present in bacteria isolated from water and street food sold in Mozambique (Doctoral dissertation, Universidade de Lisboa (Portugal)), 2024.
50. Sayson JCC, Dormido YP, Villanueva MM. Solid waste management practices of public market vendors. *B-Business Administration/Economics*, pp.66.
51. Sepadi MM, Nkosi V. Strengthening urban informal trading and improving the health of vendors: an integrated management model. *Int J Environ Res Public Health.* 2023;20(6):4836.
52. Sepadi MM, Phama LC. Street trading uncovered: a comprehensive evaluation of occupational risk and management strategies in the South African Urban Marketplace. 2024.
53. Sepadi MM. Impact of climate change on informal street vendors: a systematic review to help South Africa and Other Nations (2015–2024). *Atmosphere.* 2025;16(2):179.
54. Slootheer SL. Socio-spatial relations of food vending in Harare, Zimbabwe (Master's thesis). 2020.
55. Vertovec S, Hiebert D, Rath J. Urban markets and diversity: Toward a research agenda. 2014.
56. Zambezi FM, Muisa-Zikali N, Utete B. Effectiveness of community participation as anti-litter monitors in solid waste management in metropolitan areas in a developing country. *Environ Dev Sustain.* 2021;23:747–64.
57. Zengeni D, Muchenje B, Mbasera M. _ Nothing to show because of this dirt? 'Prospects for urban tourism under informality. *J Urban Syst Innovations Resil Zimbabwe-JUSIRZ.* 2021;3(2):172–91.

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